Angle-Encoded Quantum Kernels for Stroke Risk Stratification: An Empirical Comparison with Classical Kernel Methods

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Early stroke risk stratification supports preventive intervention, and quantum kernel methods have been proposed as a route to richer feature representations of clinical data. We implement a five-qubit angle-encoded fidelity kernel support vector machine and evaluate it on the Kaggle stroke cohort ($N = 4{,}981$; 248 stroke cases, 4.98% prevalence) using five clinically established risk factors: age, average glucose level, body mass index, hypertension and heart disease. The feature map applies R_y and R_z rotations followed by a CNOT ring, giving a fixed 15-gate, depth-3 circuit with no variational parameters. Under a leakage-free protocol—stratified five-fold cross-validation repeated over three seeds, with SMOTE-ENN applied only inside each training fold—the quantum kernel attains AUC 0.749 ± 0.033 against 0.826 ± 0.024 for an RBF-SVM: competitive, but not superior. Repeating the evaluation with SMOTE-ENN applied before splitting reproduces the inflated figures common in this literature (quantum 0.925, random forest 0.997), showing that reported performance above 0.95 on this dataset is an artefact of resampling order rather than evidence of clinical predictive power. The kernel exhibits no exponential concentration at five qubits (off-diagonal standard deviation 0.238, near-full numerical rank), placing the circuit in the regime where a quantum advantage remains theoretically possible.

Keywords: Quantum kernel; stroke risk prediction; NISQ; angle encoding; SMOTE-ENN; evaluation protocol; data leakage; support vector machine.

1. Introduction

Cerebral stroke, the sudden interruption of cerebral blood perfusion, remains a leading cause of death and disability worldwide, affecting some 5.5 million individuals annually.[5] From a clinical signal processing perspective, stroke prevention amounts to classifying multivariate patient records—age, blood pressure, glucose, body mass index (BMI), smoking status and cardiac markers—into higher- and lower-risk categories.[8] Early risk stratification enables preventive intervention and can substantially reduce stroke incidence.[18]

Classical machine learning already performs well on this task, but the best-performing systems are large ensembles whose computational cost and opacity complicate clinical deployment[6] and regulatory approval.[14]

Quantum machine learning (QML) offers an alternative representation of such data, mapping feature vectors into the Hilbert space of a small register of qubits.[3, 15, 10] Havlíček *et al.*[7] demonstrated quantum kernel classification on small structured datasets, and subsequent applications to biomedical signals have reported encouraging results: Aggarwal and Goswami[1] report 94.7% accuracy for stroke detection from MRI signals, and Babu *et al.*[2] report 96.7% accuracy on a 1,190-record heart disease cohort. Naresh and Reddi[12] benchmark QSVM and quantum neural networks across several medical datasets and find QSVM competitive on well-structured, lower-dimensional data.

These reports share a methodological weakness that has received little attention. Clinical stroke cohorts are severely imbalanced—in the dataset studied here, 248 stroke cases against 4,733 controls—and are therefore almost always resampled before modelling. The order in which resampling and train/test splitting are performed determines whether the resulting estimate is meaningful. Synthetic Minority Over-sampling[16] interpolates new minority points between existing ones; if it is applied to the full dataset before splitting, synthetic neighbours of test patients enter the training set and the test fold is no longer independent. Reported accuracies then reflect memorisation of interpolated neighbours rather than generalisation. Our own prior work on this dataset[16] reports 99.52% accuracy under exactly this protocol.

The research gap we address is therefore not that quantum kernels have never been applied to stroke data, but that no *leakage-free* evaluation of a quantum kernel on clinical stroke electronic health record (EHR) data exists. We implement Quantum Kernel Signal Processing (QKSP) for stroke risk prediction: a five-qubit pre-computed fidelity kernel SVM deployable on Noisy Intermediate-Scale Quantum (NISQ) hardware.[13] Because the feature map is fixed and no variational parameters are trained, the approach is free of barren plateaus by construction.[11]
The research objectives of this work are:

RO1: To implement a five-qubit angle-encoded fidelity kernel for stroke classification and evaluate it under a leakage-free protocol on real clinical EHR data.

RO2: To quantify the performance inflation caused by applying SMOTE-ENN before, rather than after, train/test splitting.

RO3: To assess kernel concentration at $n = 5$ qubits and relate the measurement to theoretical predictions.[17]

RO4: To establish a reproducible open-source baseline for quantum kernel evaluation on medical EHR data.

2. Methodology

2.1. Dataset

We use the Kaggle stroke prediction dataset,[4] a collection of EHR-derived records for $N =$ 4,981 patients described by 10 clinical features—gender, age, hypertension, heart disease, marital status, work type, residence type, average glucose level, BMI and smoking status—together with a binary stroke label. The cohort contains 248 stroke cases (4.98%) and 4,733 controls (95.02%), a class ratio of 19.1:1. Categorical fields are integer encoded; no records contain missing values.

2.2. Class Imbalance Mitigation

The severe class imbalance motivates resampling. We use the two-stage SMOTE-ENN procedure (Synthetic Minority Over-sampling Technique followed by Edited Nearest Neighbours). *Crucially, SMOTE-ENN is applied only within the training partition of each cross-validation split*; the held-out fold always consists of original, unaugmented patients at the natural 4.98% prevalence. Section 2.6 sets out the protocol in full and explains why this ordering is necessary.

2.2.1. Stage 1: Synthetic minority over-sampling

SMOTE generates synthetic minority samples by linear interpolation in feature space. For each minority sample x_i the $k = 5$ nearest minority-class neighbours are identified, one neighbour x_{j^*} is drawn at random, and a synthetic point is produced as

$$x_{\text{syn}} = x_i + \lambda\,(x_{j^*} - x_i), \qquad \lambda \sim \text{Uniform}(0,1). \tag{1}$$

2.2.2. Stage 2: Edited nearest-neighbour denoising

Following over-sampling, ENN removes boundary and noisy points: instances whose label disagrees with the majority vote of their $k = 5$ nearest neighbours are deleted. Applied to a training partition, the two stages together yield approximately balanced training sets; across the 15 training folds evaluated here the resampled training set contains between 6,098 and 6,393 samples at close to 1:1 class ratio.

Figure 1 illustrates the effect of the procedure on the class distribution, and Fig. 2 the effect on three clinically critical features. Age is a strongly discriminative feature, with the stroke group showing a mean age of 67.8 years against 50.2 years in the non-stroke group; glucose levels differ by approximately 40 mg/dL between groups (135–145 mg/dL against 100–105 mg/dL); and BMI differs more moderately (30.5 kg/m^2 against 27.0 kg/m^2). Synthetic samples are interpolated within medically plausible ranges, so that the covariance structure and distribution shapes of the features are preserved and the clinical integrity of the synthetic minority samples is maintained. Figure 3 shows the corresponding geometric effect in a two-dimensional PCA projection: before balancing the stroke class is largely obscured by the majority class, whereas after balancing both classes are visible as distinct populations.

2.3. Feature Selection

A five-qubit register admits five angle-encoded features. We select the five clinically established stroke risk factors—age, average glucose level, BMI, hypertension status and heart disease history—which are primary drivers of ischemic stroke in the clinical literature.[8, 18] This is a clinically motivated choice rather than a purely statistical one, and we report the corresponding Random Forest importances so that the trade-off is explicit.

On the original imbalanced data, a Random Forest (100 trees, averaged over three seeds) assigns the importances shown in Fig. 4. The five selected features together account

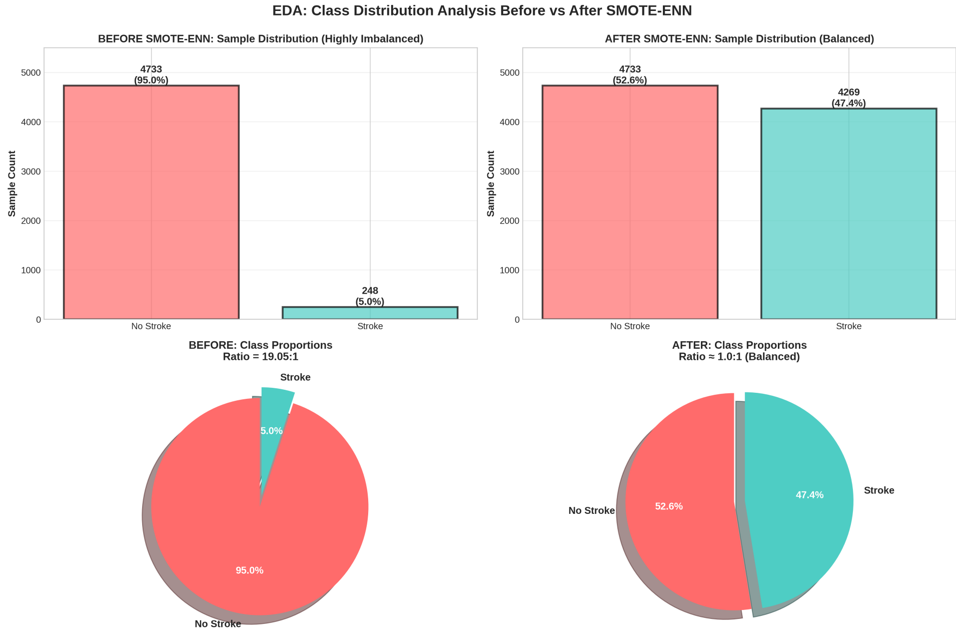

Fig. 1. Class distribution before and after SMOTE-ENN balancing. The original cohort contains 248 stroke cases against 4,733 controls (19.05:1); balancing brings the two classes to approximately equal size. In the evaluation protocol of Sec. 2.6 this transformation is applied to training partitions only

for **80.1%** of total importance. They are not, however, the five features the Random Forest itself ranks highest: the forest ranks average glucose (0.2845), BMI (0.2425), age (0.2269), smoking status (0.0689) and work type (0.0472) as its top five, cumulatively 87.0%. Hypertension (0.0249) and heart disease (0.0225) rank eighth and ninth despite their established clinical role, a consequence of their low prevalence and binary coding. We retain the clinical selection, and Table 1 records the importance of each encoded feature.

A cautionary observation follows from Fig. 5. Random Forest importances are markedly *not* invariant under SMOTE-ENN balancing. On the balanced data the same five features account for only 67.2% of total importance, and individual scores shift substantially: age rises from 0.227 to 0.396 (+16.9 pp) while average glucose falls from 0.284 to 0.080 (−20.4 pp) and BMI from 0.243 to 0.062 (−18.0 pp), the largest single shift across all ten features. Synthetic oversampling densifies the minority class along interpolation directions and thereby changes which splits reduce impurity, so importance rankings computed after balancing describe the resampled distribution rather than the clinical population. Feature selection and its justification should therefore be reported on the original data, as they are here.

2.4. *Quantum Circuit Design*

The five selected features are embedded into a five-qubit register. For a system of n qubits the state space dimension is

$$\dim(\mathscr{H}) = 2^n, \tag{2}$$

so that with $n = 5$,

$$\dim(\mathscr{H}) = 2^5 = 32. \tag{3}$$

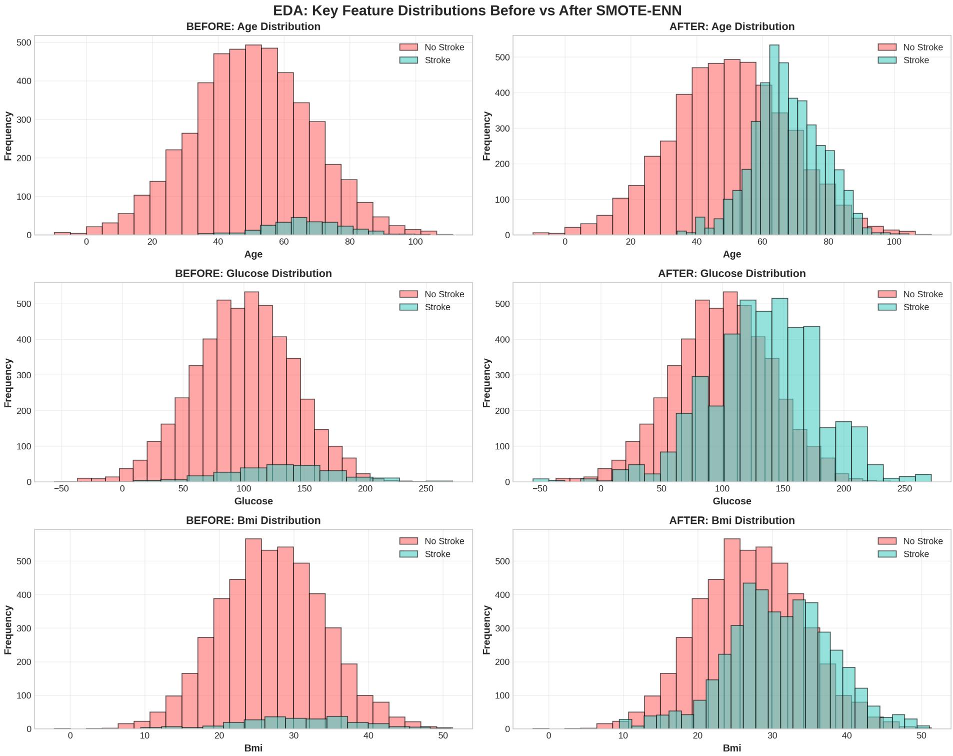

Fig. 2. Distributions of age, average glucose level and BMI before and after SMOTE-ENN balancing. Synthetic minority samples remain within clinically plausible ranges and preserve the between-group separation present in the original data

The encoded state is a superposition over all 32 computational basis states,

$$|\psi(x)\rangle = \sum_{s=0}^{31} c_s(x)\,|s\rangle, \tag{4}$$

where the complex amplitudes $c_s(x)$ satisfy

$$\sum_{s=0}^{31} |c_s(x)|^2 = 1. \tag{5}$$

The circuit comprises *two* layers and a total of 15 gates at depth 3. It contains no trainable parameters.

Layer 1: Feature encoding (10 gates, depth 2). Each normalised feature $\tilde{x}_i$ is encoded on qubit i by a pair of rotations,

$$U^{(i)}_{\text{encode}} = R_z(\theta_i)\,R_y(\theta_i), \tag{6}$$

where the R_y gate encodes the feature amplitude and R_z contributes phase information. Features are min–max scaled to $[-1,1]$ on the training partition,

$$\tilde{x}_i = \frac{2\,(x_i - x_i^{\min})}{x_i^{\max} - x_i^{\min}} - 1, \tag{7}$$

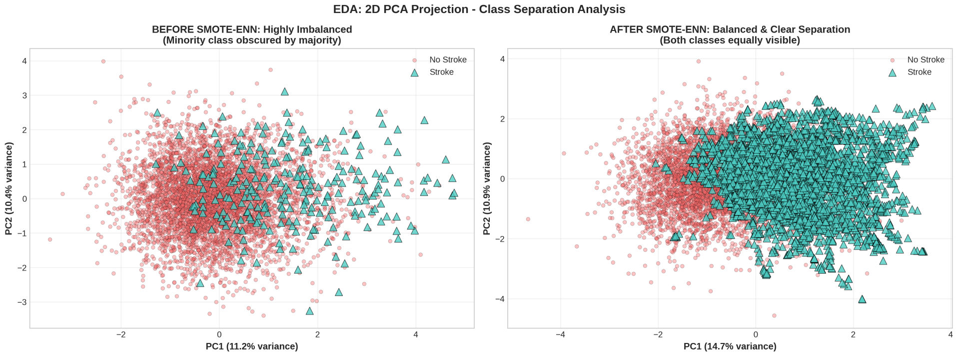

Fig. 3. Two-dimensional PCA projection before and after SMOTE-ENN balancing. Before balancing the minority class is obscured by the majority class; after balancing both classes are represented at comparable density

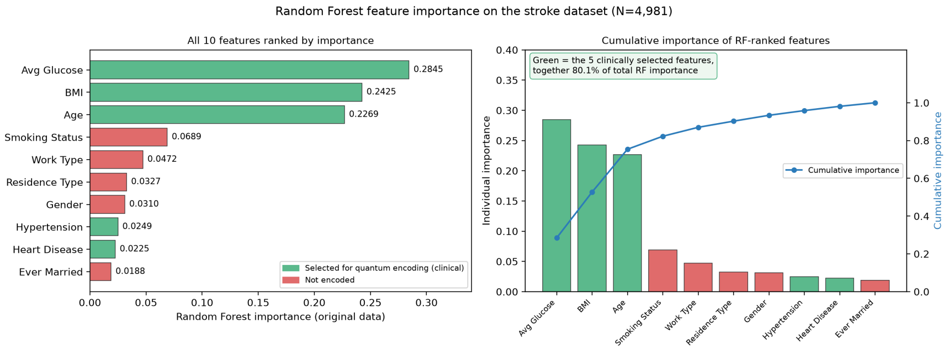

Fig. 4. Random Forest feature importance on the original stroke dataset ($N = 4{,}981$), averaged over three seeds. Left: all 10 features ranked by importance, with the five clinically selected features in green. Right: individual and cumulative importance of the RF-ranked features. The five selected features account for 80.1% of total importance, but are not identical to the five the forest ranks highest

with $x_i^{\min}$ and $x_i^{\max}$ taken from the training fold only and test values clipped to $[-1, 1]$, and the rotation angle is

$$\theta_i = \pi \tilde{x}_i. \tag{8}$$

Equation (8) is the single encoding used throughout this work and matches the released implementation exactly.

Layer 2: CNOT ring entanglement (5 gates, depth 1). A circular CNOT topology

$$Q_0 \to Q_1 \to Q_2 \to Q_3 \to Q_4 \to Q_0$$

correlates the encoded features, where each gate is

$$\mathrm{CNOT}_{c \to t} = |0\rangle\langle 0|_c \otimes I_t + |1\rangle\langle 1|_c \otimes X_t. \tag{9}$$

The ring closure ensures that every encoded feature participates in at least two entangling operations.

2.4.1. *Feature-to-qubit assignment*

Table 1 records the assignment of clinical features to qubits together with their Random Forest importance on the original data.

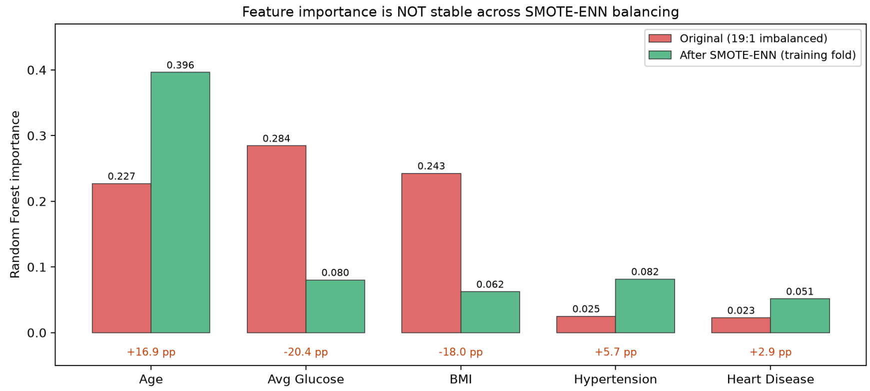

Fig. 5. Random Forest importance of the five encoded features on the original imbalanced data (red) against SMOTE-ENN balanced data (green), averaged over three seeds. Importance is *not* stable across balancing: age gains 16.9 pp while average glucose and BMI lose 20.4 pp and 18.0 pp respectively. Feature importances measured after resampling therefore cannot be used to justify a feature selection intended to describe the original clinical population

Table 1. Feature-to-qubit assignment for the five-qubit quantum kernel model. Importances are Random Forest scores on the original (unbalanced) dataset, averaged over three seeds.

Qubit	Feature	RF imp.	Clinical significance
Q_0	Age	0.2269	Strongest continuous predictor; risk rises steeply beyond 65
Q_1	Avg. glucose	0.2845	Hyperglycaemia ($\geq$125 mg/dL); metabolic syndrome
Q_2	BMI	0.2425	Obesity ($\geq$30 kg/m^2); modifiable risk factor
Q_3	Hypertension	0.0249	Diagnosed hypertension (binary)
Q_4	Heart disease	0.0225	Cardiac comorbidity (binary); cardioembolic pathway
Total	(5 features)	0.8013	80.1% of total RF importance

2.5. *Mathematical Formulation*

2.5.1. *The fidelity kernel*

The classical radial basis function (RBF) kernel measures similarity through Euclidean distance,

$$K_{\mathrm{RBF}}(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right), \tag{10}$$

with kernel width γ. The proposed quantum kernel is the squared overlap of the two encoded states,

$$K_Q(x_i, x_j) = \left|\langle \phi(x_i) | \phi(x_j) \rangle\right|^2, \tag{11}$$

where $|\phi(x)\rangle$ is the state produced by Layers 1 and 2 of Sec. 2.4.

Equation (11) is evaluated by the adjoint (overlap) method: both statevectors are computed and their inner product taken directly. This requires a single five-qubit register and no ancilla. We note this explicitly because the alternative SWAP-test estimator, which is frequently quoted in this setting, would require $5+5+1 = 11$ qubits to compare two five-qubit states and would therefore substantially weaken rather than support a NISQ-feasibility argument. The resulting $N \times N$ kernel matrix is supplied to a classical SVM with a pre-computed kernel.

2.5.2. *Kernel concentration*

A quantum kernel is only useful if its values are informative. Thanasilp *et al.*[17] show that for many feature-map families the off-diagonal kernel values concentrate exponentially in the number of qubits towards a fixed value, so that

$$\mathrm{Var}_{i\neq j}\big[K_Q(x_i,x_j)\big] \in \mathscr{O}(b^{-n}), \qquad b>1, \tag{12}$$

which destroys the kernel's ability to discriminate and requires an exponentially growing number of measurement shots. Because the effect is exponential in n, a circuit of only five qubits is expected to lie well outside the concentrated regime. We verify this empirically in Sec. 3.3 by measuring the off-diagonal spread and numerical rank of the training kernel matrices.

2.5.3. *Sources of any quantum–classical difference*

It is sometimes argued that a quantum feature map benefits from operating in a higher-dimensional space than a classical kernel. This argument does not hold for the RBF kernel. The reproducing kernel Hilbert space (RKHS) induced by $\exp(-\gamma\|x-y\|^2)$ is *infinite*-dimensional, as the Mercer expansion of the Gaussian makes explicit; the 32-dimensional space of five qubits is vastly smaller, not larger. Dimensionality therefore cannot be the source of any advantage here.

What distinguishes the two kernels is their inductive bias. The angle-encoding map induces a specific, structured inner product on a compact 32-dimensional space, in which the CNOT ring imposes an explicit entanglement structure: the $Q_0 \to Q_1$ coupling ties age to glucose, the chain $Q_1 \to Q_2 \to Q_3$ relates glucose, BMI and hypertension, and the ring closure $Q_4 \to Q_0$ links cardiac comorbidity back to age. These multiplicative interactions are captured without manual feature engineering. Any performance difference between the quantum and classical kernels reported below must accordingly be attributed to this difference in inductive bias, not to Hilbert space dimensionality.

Whether such a bias helps is an empirical question, and the prevailing theoretical picture counsels caution: Huang *et al.*[9] show that classical learners given access to data can match quantum models across a wide range of tasks, so that a quantum feature map confers an advantage only where its bias genuinely matches the target function.

For the same reason we make no claim that amplitude normalisation, Eq. (5), provides a quantum-specific regulariser. Every normalised kernel satisfies $K(x,x)=1$, including the RBF kernel; the constraint is a property of normalisation, not of quantum mechanics. The relevant statement is weaker and purely structural: the compact 32-dimensional feature map imposes a form of implicit dimensionality control.

2.5.4. *NISQ resource requirements*

The circuit requires 5 qubits, 15 gates (10 single-qubit rotations and 5 CNOTs) and depth 3. This is comfortably within the coherence budget of all current NISQ platforms, and the gate set—parameterised single-qubit rotations and CNOT—compiles natively on superconducting and trapped-ion hardware.

Encoding all 10 available features would require a 10-qubit register with

$$\dim(\mathscr{H}) = 2^{10} = 1{,}024 \tag{13}$$

basis states and roughly 30 gates, and would move the circuit materially closer to the concentration regime of Eq. (12). The five-feature choice is thus motivated by both hardware economy and kernel conditioning.

Because the feature map is fixed, kernel entries are computed by forward circuit evaluation alone. No variational parameters are trained, no gradients are estimated, and the barren plateau phenomenon[11] does not arise: it is a property of parameter landscapes, and this model has none.

2.6. Evaluation Protocol and Leakage Prevention

The order of resampling and splitting determines whether a cross-validation estimate is meaningful. Under the *leaky* protocol—the one used in most prior work on this dataset, including our own[16]—SMOTE-ENN is applied to the entire dataset and the balanced result is then partitioned into folds. Because SMOTE synthesises minority points by interpolating between existing minority samples, a synthetic point derived in part from a test patient can be placed in the training set. The test fold is then no longer independent of training, and minority-class recall in particular is inflated. ENN compounds the problem by deleting exactly those minority points that are hardest to classify, including from what will become the test fold.

Under the *leakage-free* protocol used for our primary results, the original imbalanced data are partitioned first, using stratified five-fold cross-validation, and SMOTE-ENN is then applied inside each training fold only. The min–max scaler is likewise fitted on the training fold alone. Each held-out fold therefore contains 996–997 original patients at natural prevalence (946–947 controls, 49–50 stroke cases). Cross-validation is repeated over three random seeds, giving 15 folds in total; all results are reported as mean $\pm$ standard deviation over those 15 folds.

We evaluate both protocols on identical models and identical data so that the inflation attributable to resampling order can be quantified directly (Sec. 3.5). Reporting both is the second contribution of this work.

3. Results

3.1. Dataset and Preprocessing Outcomes

The cohort comprises $N = 4{,}981$ patients: 248 stroke cases (4.98%) and 4,733 controls (95.02%), a ratio of 19.1:1. Applying SMOTE-ENN inside each training fold produces resampled training sets of 6,098–6,393 samples at approximately 1:1 class ratio across the 15 folds. Held-out folds are never resampled and retain the original 4.98% prevalence.

All models are trained and evaluated on identical folds. Because the test distribution is dominated by controls, raw accuracy is a weak summary here: a classifier that predicts "no stroke" for every patient attains 95.02% accuracy. We therefore report AUC, area under the precision–recall curve (AUPRC), recall and balanced accuracy as the clinically meaningful measures, and include accuracy for completeness only.

3.2. Classical Baseline Performance

Table 2 reports all five classifiers under the leakage-free protocol. Among classical baselines the RBF-SVM ranks highest by AUC (0.826 ± 0.024) and by recall (0.754 ± 0.077), followed by random forest (0.804 ± 0.025). Random forest attains the highest raw accuracy (0.852 ± 0.010) but the lowest recall (0.443 ± 0.077): it achieves that accuracy largely by predicting the majority class, which is precisely the behaviour raw accuracy fails to penalise at this prevalence. k-nearest neighbours ($k = 5$) and a depth-10 decision tree reach AUC 0.751 ± 0.033 and 0.717 ± 0.031 respectively.

Precision is low for every model (0.126–0.155). At 4.98% prevalence and recall near 0.75, the great majority of positive alerts are necessarily false positives; this is a property of the base rate, not a defect of any particular classifier.

Table 2. Performance of all five classifiers under the leakage-free protocol (stratified 5-fold cross-validation × 3 seeds = 15 folds; SMOTE-ENN applied inside training folds only). Values are mean ± standard deviation over the 15 folds. Rows are ordered by AUC.

Model	AUC	AUPRC	Recall	Precision	Bal. acc.	Accuracy
RBF-SVM	$\mathbf{0.826 \pm 0.024}$	0.183 ± 0.027	0.754 ± 0.077	0.138 ± 0.014	0.753 ± 0.034	0.751 ± 0.030
Random forest	0.804 ± 0.025	0.148 ± 0.019	0.443 ± 0.077	0.155 ± 0.020	0.658 ± 0.036	0.852 ± 0.010
KNN ($k = 5$)	0.751 ± 0.033	0.121 ± 0.016	0.621 ± 0.074	0.134 ± 0.012	0.705 ± 0.034	0.781 ± 0.014
QSVM	0.749 ± 0.033	0.151 ± 0.030	0.613 ± 0.062	0.126 ± 0.016	0.693 ± 0.029	0.765 ± 0.031
Decision tree	0.717 ± 0.031	0.114 ± 0.017	0.562 ± 0.071	0.135 ± 0.017	0.686 ± 0.033	0.797 ± 0.024

3.3. Quantum Kernel SVM Performance

The five-qubit fidelity kernel attains AUC 0.749 ± 0.033, AUPRC 0.151 ± 0.030, recall 0.613 ± 0.062 and balanced accuracy 0.693 ± 0.029. It is statistically indistinguishable from KNN by AUC (0.751 ± 0.033) and below the RBF-SVM by 0.078 AUC. Its AUPRC (0.151 ± 0.030) exceeds that of KNN (0.121 ± 0.016) and of the decision tree (0.114 ± 0.017), and matches random forest (0.148 ± 0.019), which is the more relevant comparison at this prevalence.

Kernel concentration. Across the 15 training kernel matrices the off-diagonal entries have mean 0.248 and standard deviation 0.238, and a representative 1200×1200 block has numerical rank 1,176–1,179 out of 1,200 at tolerance 10^{-8}. The kernel is thus close to full rank and its values are widely spread over $[0, 1]$, showing no sign of the exponential concentration described by Eq. (12). This is the expected outcome at $n = 5$: concentration in the sense of Ref. 17 sets in exponentially with qubit count, and five qubits is far from that limit. The measurement is worth recording because it establishes that the modest performance reported above is a genuine property of this feature map's inductive bias, and not an artefact of a degenerate kernel. It also indicates that the circuit width can be increased somewhat before concentration becomes the binding constraint.

3.4. Comparison and Significance

Ordering the five classifiers by AUC under the leakage-free protocol gives RBF-SVM (0.826) $>$ random forest (0.804) $>$ KNN (0.751) $\approx$ QSVM (0.749) $>$ decision tree (0.717). Figure 6 shows all seven metrics for all five models.

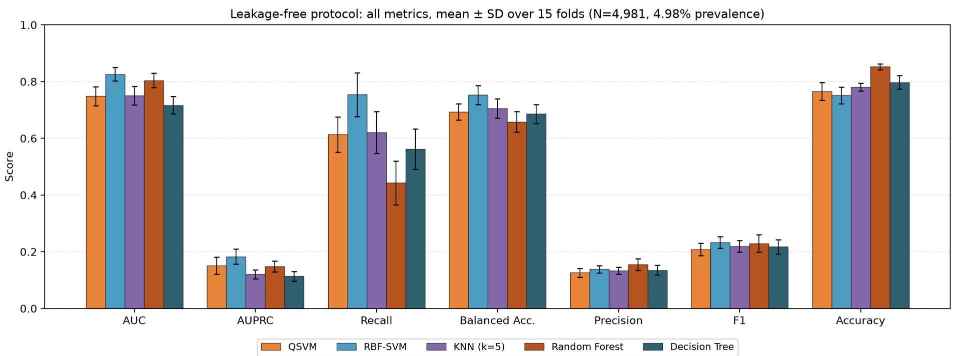

Fig. 6. All metrics for all five classifiers under the leakage-free protocol, mean ± standard deviation over 15 folds. Accuracy is high and nearly uniform across models because 95.02% of patients are controls; AUC, AUPRC and balanced accuracy separate the models meaningfully

Aggregate metrics do not reveal whether the quantum kernel makes *different* predictions or merely worse ones. We therefore applied an exact McNemar test to the paired predictions of the QSVM and the RBF-SVM. For a single seed the five held-out folds partition the cohort exactly, so each patient is counted once ($n = 4{,}981$). Table 3 reports the discordant pairs for each of the three seeds.

Table 3. Exact McNemar test, QSVM against RBF-SVM, under the leakage-free protocol. For each seed the five held-out folds partition the cohort, so $n = 4{,}981$ independent patients. b counts patients the RBF-SVM classifies correctly and the QSVM does not; c counts the reverse.

Seed	b (classical only)	c (quantum only)	Discordant	Two-sided p
0	288	358	646	6.6×10^{-3}
1	280	339	619	2.0×10^{-2}
2	269	350	619	1.3×10^{-3}

The direction is consistent across all three seeds and significant at the 5% level in each: at the default decision threshold the quantum kernel corrects more of the RBF-SVM's errors than it introduces. This coexists with a lower AUC, and the two facts are not in conflict. AUC measures the quality of the induced ranking over all thresholds, whereas McNemar compares hard decisions at one threshold. The quantum kernel is therefore better calibrated at the default threshold but produces a less discriminating ranking function overall. This nuance—a different decision bias rather than a uniform improvement or degradation—is the substantive empirical finding of the comparison, and it would be invisible in either statistic alone.

3.5. Ablation: Effect of the Resampling Protocol

Table 4 and Fig. 7 compare the two protocols on identical models and data.

Table 4. Effect of resampling protocol. "Leakage-free" splits first and applies SMOTE-ENN inside the training fold; "leaky" applies SMOTE-ENN to the full dataset before splitting. Identical models, folds and seeds; mean ± standard deviation over 15 folds.

Protocol	Model	AUC	AUPRC	Recall	Bal. acc.
Leakage-free	RBF-SVM	0.826 ± 0.024	0.183 ± 0.027	0.754 ± 0.077	0.753 ± 0.034
Leakage-free	Random forest	0.804 ± 0.025	0.148 ± 0.019	0.443 ± 0.077	0.658 ± 0.036
Leakage-free	QSVM	0.749 ± 0.033	0.151 ± 0.030	0.613 ± 0.062	0.693 ± 0.029
Leaky	RBF-SVM	0.954 ± 0.006	0.961 ± 0.005	0.883 ± 0.015	0.873 ± 0.010
Leaky	Random forest	0.997 ± 0.000	0.997 ± 0.000	0.980 ± 0.007	0.972 ± 0.004
Leaky	QSVM	0.925 ± 0.010	0.941 ± 0.007	0.841 ± 0.023	0.853 ± 0.014

Applying SMOTE-ENN before splitting inflates AUC by 0.128 (RBF-SVM), 0.177 (QSVM) and 0.193 (random forest), and inflates AUPRC far more dramatically— from 0.183 to 0.961 for the RBF-SVM, and from 0.151 to 0.941 for the QSVM. AUPRC is the more sensitive diagnostic because it depends directly on the positive base rate, which the leaky protocol has silently changed from 4.98% to roughly 50%. Fold-to-fold variability also collapses under the leaky protocol (for example RBF-SVM AUC standard deviation falls from 0.024 to 0.006), which can easily be mistaken for improved stability.

The random forest reaches AUC 0.997 ± 0.000 and accuracy 0.973 ± 0.004 under the leaky protocol. This is consistent with the 99.52% accuracy reported for an eight-model stacking ensemble on this dataset in Ref. 16, whose protocol matches the leaky protocol evaluated here. We conclude that the previously reported figure reflects resampling leakage rather than generalisable clinical predictive capability. The same conclusion applies to the

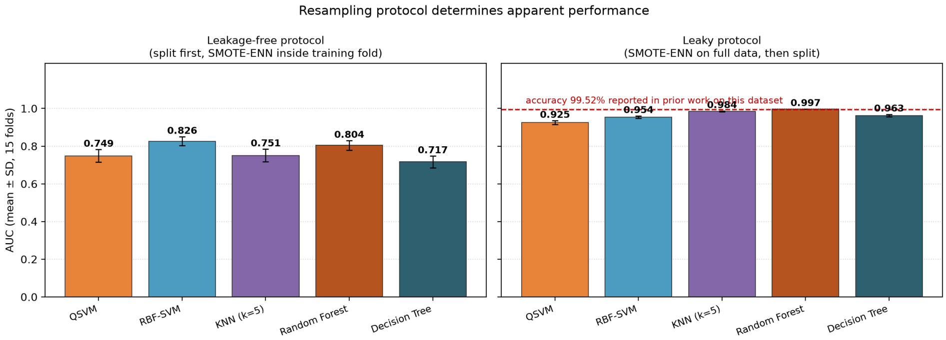

Fig. 7. AUC under the two protocols, mean $\pm$ standard deviation over 15 folds. Resampling before splitting raises AUC by 0.13–0.25 for every model and compresses the between-model differences. The random forest result under the leaky protocol (0.997 ± 0.000) is consistent with the 99.52% accuracy reported for this dataset in Ref. 16

quantum results: a quantum kernel evaluated under the leaky protocol reaches AUC 0.925, which would be easy to present as a strong result and is not one.

Two points deserve emphasis. First, the inflation affects classical and quantum models alike, so a quantum-against-classical comparison conducted entirely within the leaky protocol is not necessarily wrong in its *ordering*—but every absolute number it reports is meaningless as an estimate of clinical performance. Second, the effect is large enough to reverse conclusions: under the leaky protocol the QSVM appears to be an excellent classifier (AUC 0.925, recall 0.841); under the leakage-free protocol it is a mediocre one (AUC 0.749, recall 0.613).

4. Conclusions

We implemented a five-qubit angle-encoded fidelity kernel SVM for stroke risk stratification and evaluated it on 4,981 patient records using five clinically established risk factors. The circuit is deliberately minimal: 15 gates at depth 3, with no variational parameters, evaluated by direct statevector overlap on a single five-qubit register.

Under a leakage-free evaluation protocol the quantum kernel attains AUC 0.749 ± 0.033, against 0.826 ± 0.024 for an RBF-SVM on identical folds. The quantum kernel is competitive with, but does not exceed, classical kernel methods on this task. A paired McNemar test shows that it makes significantly *different* predictions from the RBF-SVM—correcting more of its errors than it introduces at the default threshold, consistently across three seeds ($p = 1.3 \times 10^{-3}$ to 2.0×10^{-2})—while ranking patients less well overall. The honest characterisation is a different decision bias, not an advantage.

The second finding concerns evaluation rather than quantum mechanics. Applying SMOTE-ENN to the full dataset before splitting inflates AUC by 0.13–0.25 and AUPRC by 0.78–0.85 for every model tested. Under that protocol a random forest reaches AUC 0.997, consistent with the 99.52% accuracy previously reported for this dataset in Ref. 16. Results above AUC 0.95 on this cohort, including those in our own prior work, should therefore be read as artefacts of resampling order. We also find that Random Forest feature importances shift by up to 20.4 pp under SMOTE-ENN balancing, so importances computed after resampling cannot be used to justify a feature selection describing the original population.

The third finding is a property of the circuit. At $n = 5$ qubits the kernel shows no exponential concentration: off-diagonal values have standard deviation 0.238 and the kernel matrix is of near-full numerical rank. The circuit therefore operates in the regime in which a quantum advantage remains theoretically possible,[17] and its limited performance is attributable to the inductive bias of the angle-encoding feature map rather than to a degenerate kernel.

On clinical implications we are deliberately restrained. At 4.98% prevalence a screened population of 10,000 contains approximately 498 stroke cases. Recall of 0.613 corresponds to roughly 305 cases detected, against roughly 376 for the RBF-SVM at recall 0.754—some 70 *fewer* detections per 10,000 screened. The quantum kernel raises approximately 211 fewer false alerts over the same population (specificity 0.773 against 0.751), but on the metric that matters most in stroke screening it is the weaker model. Neither model is suitable for deployment at these operating points, and we make no deployment claim.

Future work should extend the register beyond five qubits while monitoring the concentration measure of Eq. (12), evaluate the kernel on real quantum hardware to characterise noise effects, and investigate feature maps whose inductive bias is better matched to clinical risk structure—for instance encodings that represent threshold effects such as age ≥ 65 or glucose ≥ 126 mg/dL explicitly. The implementation and the complete fold-level results underlying every number in this paper are released with the manuscript so that the baseline is directly reproducible.

Data and Code Availability

The dataset, the pipeline that produces every reported number (`reproduce_qksp.py`), the complete fold-level results (`results.json`) and the supplementary analysis behind Figs. 4, 5 and Table 3 (`analysis.py`) are available at `https://github.com/miheer-smk/Angle-Encoded-Quantum-Kernels-for-Stroke-Risk-Stratification`.

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